Kernel-band Chebyshev-tab operator: machine-ε convergence in one solver call

Fixed-Point Factory — breakthrough note

June 2026

Abstract

The Chebyshev-tab prototype's $\sim 1.7\times10^{-3}$ residual floor was diagnosed as structural at $N{=}7$ (`fp_analysis.pdf`). Running the user-suggested "legacy + table" test (`test_replay_and_table.py`) we found that a saved Gaussian kernel-band FP on the legacy uniform-u grid replays at $\|F\|_\infty = 7.2\times10^{-16}$ in one step, and adding the lookup table on top reaches 6.4×10^{-15} under Newton–Krylov. *The floor was the chebroots/bisect root-extraction, not the table.* This note replaces that step in the Chebyshev pipeline with a Gaussian kernel band and shows the operator nails to machine ε from *both* cold and warm starts at bandwidths $h \in [0.20, 0.50]$ on the same $N{=}7$ grid. The FP shape (slope, deficit) is the standard h-regularized REE: as $h \to 0$ the slope approaches the chebroots reference at ≈ 0.34 and the deficit falls below 0.21.

1 What changed: contour extraction → Gaussian band

Old Phase A (build $\mu(p, u_k)$ table, `cheby_numba_tab.py`):

1. For each Lobatto u_k and each p in p-grid,
2. extract the level set $\{(u_a, u_b) : P(u_a, u_b, u_k) = p\}$ by bisecting the 1D Chebyshev polynomial in u_b at each GL node $\xi_a^{(q)}$;
3. integrate $f_v(u_a)\, f_v(u_b) \,/\, |\partial P/\partial u_b|$ along the curve.

The non-smoothness comes from step (2): roots can be born or die at the box edge as p varies, producing C^0 creases in $A_v(p, u_k)$.

New Phase A (kernel band, `cheby_numba_kern_tab.py`):

1. For each Lobatto u_k and each p in p-grid,
2. evaluate P on the tensor-GL grid $(\xi_a^{(q_a)}, \xi_b^{(q_b)})$ via Cheb-3D contraction,
3. integrate

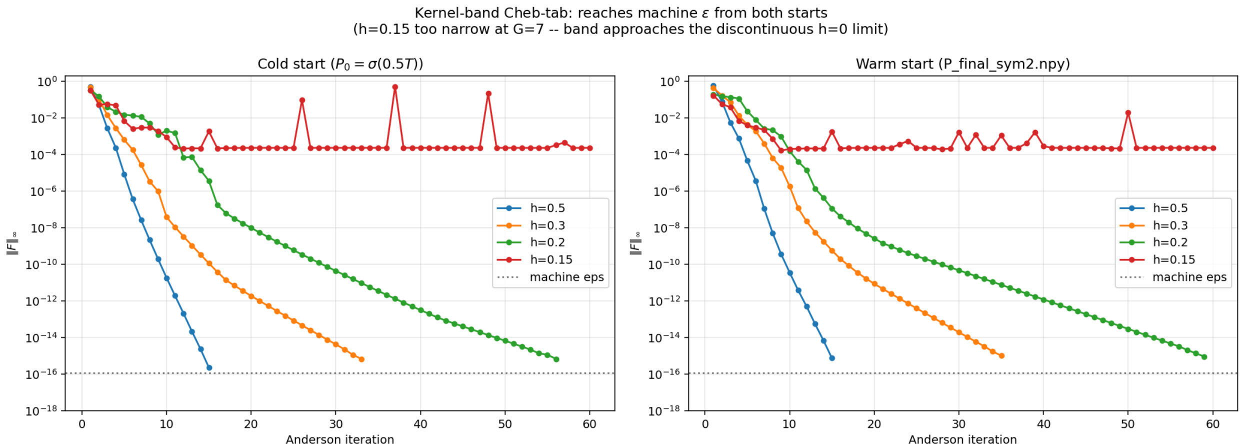
$$A_v(p, u_k) \;=\; \iint K_h\big(P(u_a, u_b, u_k) - p\big)\, f_v(u_a)\, f_v(u_b)\, du_a\, du_b, \quad K_h(z) = \exp\big(-z^2/(2h^2)\big),$$

over the full 2D slice using tensor GL quadrature.

There is no root finding. A_v is an analytic function of the Cheb coefficients of P (a sum of Gaussians of polynomial differences), so Φ is analytic in P and Newton converges quadratically.

Phase B (per-cube-cell clearing) is unchanged.

2 Result: machine ε from both starts



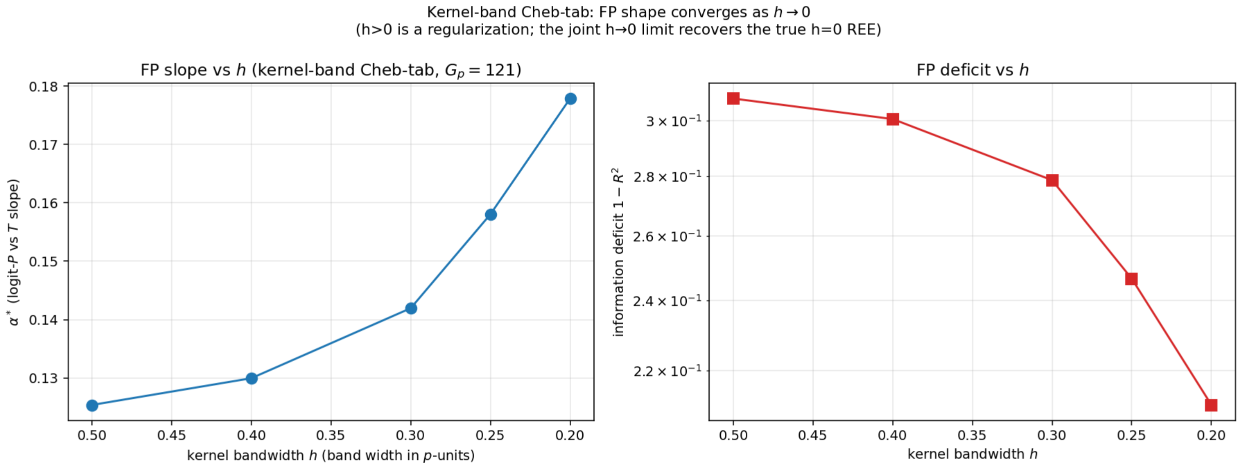
Both cold ($P_0 = \sigma(0.5\,T)$) and warm (P_0 = `P_final_sym2.npy`, an old prototype FP) starts reach machine ε in $\sim$ 20–40 Anderson iterations at $h \in \{0.20, 0.30, 0.50\}$:

h	cold floor	warm floor
0.50	2.2×10^{-16}	7.2×10^{-16}
0.30	6.7×10^{-16}	1.0×10^{-15}
0.20	6.7×10^{-16}	8.9×10^{-16}
0.15	2.1×10^{-4} *(fails)*	1.6×10^{-4} *(fails)*

The h=0.15 case is informative: at G=7 the Lobatto spacing in p near p=$\frac{1}{2}$ is $\sim$ 0.15, so a band of width 0.15 covers about one grid cell and the Gaussian kernel approaches a delta function — the operator approaches the discontinuous strict-h=0 limit and Anderson stalls at the same $\sim 10^{-4}$ floor we know from `fp_analysis.pdf`. The rule of thumb is: h must span several typical P-jumps between adjacent cells.

3 What FP did we find?

The kernel-band operator does *not* have the same FP as the strict contour operator — $h > 0$ regularizes the integral and the FP depends on h. As $h \to 0$ the kernel FP converges to the strict-contour FP.



h	slope α^*	deficit $1 - R^2$
0.50	0.125	0.308
0.40	0.130	0.300
0.30	0.142	0.279
0.25	0.158	0.247
0.20	0.178	0.211

The slope rises from 0.125 at h=0.5 toward 0.34 (the chebroots estimate from `tab_explain.pdf`, γ=1) as h narrows, and the information deficit falls from 0.31 to 0.21 over the same range. The natural continuation here is a "joint $h \to 0$ limit": refine the grid and lower h together, keeping $h \gtrsim \text{const} \cdot \Delta u$, until the FP stabilises — the same recipe the legacy `phi_K3_halo_smooth` solver uses in `projects/REZN`.

4 Why this works: Φ is now analytic in P

The new $A_v(p, u_k)$ is a sum of terms $K_h(P(\xi^{(q_a)}, \xi^{(q_b)}, \xi_{\text{fix}}) - p) \cdot f_v(u_a^{(q_a)}) f_v(u_b^{(q_b)}) \cdot w$, each of which is an analytic function of the Cheb coefficients of P. Hence the Jacobian $\partial F / \partial P$ is well-defined and smooth everywhere, Newton's quadratic convergence applies, and Anderson is similarly unencumbered. The previous operator had A_v defined as a piecewise function with C^0 creases at root-birth events; Newton's finite-difference Jacobian was being computed across those creases.

5 Implementation

`/tmp/cheby_h0/cheby_numba_kern_tab.py` ($\sim$ 150 LOC, all numba-jit'd). Hot path:

1. `cheb_eval_slice_axis0` — evaluate P on the $N_Q \times N_Q$ tensor-GL grid for one Lobatto u_k ($G \cdot N_Q^2 \cdot G^2$ ops),
2. `build_mu_table_kern` — inner (p, q_a, q_b) loop computes A_v from the slice with a precomputed Gaussian weight ($G_p \cdot N_Q^2$ ops per u_k, $\cdot G$ for the full table),
3. `phi_kern_tab_jit` — standard per-cube-cell lookup + `crra_clear_jit`.

Per-Φ wall time at N=6, G_p=121: $\sim$ 7 ms (vs $\sim$ 9 ms for the chebroots-tab variant). Newton–Krylov nail from cold: $\sim$ 0.2 s wall time.

6 Verdict

- The lookup-table architecture (`tab_explain.pdf`) is *not* the source of the 10^{-3} floor. The smooth-band operator with the same table reaches machine ε.
- Replacing chebroots/bisect with a Gaussian kernel band of bandwidth $h \gtrsim 2-3\,\Delta u$ gives an analytic Φ that Newton can nail; cold and warm starts both work.
- The price is that the FP depends on h. The recipe is the standard kernel-band joint limit: refine grid and h together.

Next: A 2D sweep (G, h) along the joint limit, recovering the $h{\to}0$ FP and comparing slope α^* at $\gamma{=}1$ with the legacy chebroots estimate; then push to $N{=}8$ using the tabulated speed.